

KAMESH BALAJI

Associate Engineer - Backend Developer

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🌐 [LinkedIn](#) [Portfolio](#) [Github](#)

Professional Experience

Aravind Eye Care System - Aravind Eye Hospital

Associate Engineer - IT
IT Intern

June 2024 – Present
February 2024 – May 2024

Core Competencies

Languages: Python, SQL, C#
Backend Framework: ASP.NET Core
Databases: MS SQL Server, PostgreSQL
BI and Reporting Tools: Power BI, Fast Report.NET
Data Engineering Tools : Apache Airflow, Apache Kafka and Minio
Containerization and Orchestration Tools: Docker, Kubernetes (Microk8s)
SCM and CICD tools: Git, Github and Jenkins
Web Servers : Nginx and Internet Information Services (IIS)
Monitoring Tools: Loki, Prometheus and Grafana

Project Portfolio

Web Api - Based Clinical Reporting for EMR Systems

- **Tools and Technologies :** ASP DotNet Core 8.0 (REST Api, Clean Architecture), PostgreSQL and SQL Server, Fast Report .NET
- Developed a **.NET 8 Web API microservice** leveraging both **Dapper ORM** and **ADO.NET** for efficient data access and integration of patient records from **PostgreSQL** and **SQL Server** databases.
- Applied **Clean Architecture** principles to enforce separation of concerns and ensure a scalable, testable, and maintainable codebase.
- Implemented dynamic and customizable healthcare reports using the **FastReport** package, supporting export in PDF format

SSRS-Based Clinical Reporting for EMR Systems

- **Tools and Technologies :** SQL Server Reporting Services, SQL Server, Microsoft PowerBI, ASP DotNet Framework MVC
- Configured and deployed patient summary reports to **SSRS Report Server 2017**, enabling server-side report management
- Retrieved reports in **PDF format** by making authenticated **HTTP GET** requests to SSRS **report execution endpoints** using **.NET Framework** code
- Integrated SSRS report rendering into an **ASP.NET MVC** web application using **WebClient / HttpClient** for dynamic report generation
- Fetched patient data from **SQL Server** and **PostgreSQL** using stored procedures and parameterized queries as report data sources

Innovative Solutions and POCs

- **Report Development with .NET Web API Integration**
 - **Tech Stack:** .NET Core, Power Bi Report Builder
 - **Challenge:** Generating RDL reports using SSRS was time-consuming and inflexible for real-time analytics.
 - **Solution:** Implemented dynamic report generation within the .NET Core Web API by embedding reporting logic, optimizing SQL queries, and enabling export in multiple formats. This improved latency and flexibility for on-demand reporting.
- **YARP Reverse Proxy with Cryptographic Payload Encryption for Secure Communication**
 - **Tech Stack:** .NET Core, Angular, AES Symmetric Encryption
 - **Challenge:** Sensitive EMR data exposed during Web API transmissions was vulnerable to breaches and tampering.
 - **Solution:** Designed a POC to encrypt and decrypt API payloads using AES at Reverse Proxy Layer. Ensured end-to-end secure communication between the client (Angular) and server (.NET Core), mitigating risks of data exposure and manipulation.
- **Kubernetes and System Monitoring with Prometheus and Grafana**
 - **Tech Stack:** Prometheus, Grafana, Microk8s, Node Exporter, Windows Exporter
 - **Objective:** Enable real-time monitoring of Kubernetes clusters and host systems.
 - **Implementation:**
 - * Deployed Prometheus and Grafana in Microk8s to monitor cluster metrics.
 - * Integrated Node Exporter and Windows Exporter to collect Linux and Windows host metrics.
 - * Configured Grafana dashboards for system health, resource usage, and cluster activity.
 - * Set up email alerting in Grafana to notify on threshold breaches and abnormal behaviors.
- **User Interaction Analytics with Matomo and Angular**
 - **Tools and Technologies:** Matomo, NGINX, Angular
 - Deployed and configured the open-source **Matomo analytics platform** on an NGINX web server, enabling a privacy-focused alternative to Google Analytics.
 - Tuned NGINX settings for optimal performance, including HTTPS setup, caching, and secure access to the Matomo dashboard.
 - Integrated **Matomo** tracking into a modern **Angular** application by configuring the **Matomo JavaScript tracker** and custom event tracking.

Publications And Awards

Above and Beyond Award

AUROiTECH - Annual Day 2025

Recognized for outstanding contributions and dedication

- Received the "Above and Beyond Award" from **AUROiTech - Aravind Eye Care System** in recognition of exceptional performance and contributions at the Annual Day 2025 celebration.

Conference Publication

ICSSS'23

Supporting system for hemiplegic patients with health monitoring capabilities

- Published a paper titled "Supportive System for Hemiplegic Patients with Health Monitoring Capabilities" at ICSSS'23 conference, indexed in IEEE Xplore with Scopus.

Academic Profile

Saveetha Engineering College

2020 - 2024

B.E in Biomedical Engineering

90 Percentage

Annai Sarada Matric. Hr. Sec. School

2020

H.S.C

84.6 Percentage

Annai Sarada Matric. Hr. Sec. School

2018

S.S.L.C

86.4 Percentage